

insight

#3

Purpose first: The need for a paradigm shift in privacy-preserving data sharing

**Dr Theresa Stadler,
Prof. Carmela Troncoso
& Dr Melanie Kolbe-Guyot**

Executive summary

Data sharing and data publishing have become important subjects on the European political agenda. Current discussions are focused on how to make commercial, research, and public data, particularly microdata, available to a broader set of actors while preserving the privacy of affected individuals. Privacy-enhancing technologies (PETs) for data sharing are considered a key solution for achieving this goal. These technologies are often promoted as “silver bullet” solutions that can address the long-standing technical challenges of ensuring data privacy while maintaining high data utility. The recent enthusiasm for synthetic data generation (SDG) methods serves as a notable example of this trend.

This policy paper highlights three concerns with this current trend. First, data-sharing policies and guidelines rarely acknowledge that there is an inherent fundamental trade-off between ensuring privacy and preserving data utility - a trade-off which can be minimized but never completely resolved. This trade-off is especially pronounced in data-sharing scenarios that require high flexibility in analysis tasks, where the shared data must retain its utility for potentially “all-purpose” use cases.

Second, the frequent recommendations about specific privacy technologies for data sharing from public, research, and private entities often do not adequately address the empirical evidence that challenges these technologies’ purported effectiveness. Third, as companies increasingly label certain technologies as “privacy-preserving,” there is a risk of creating power imbalances in the data supply chain. Using these technologies safely requires expertise that many data profes-

sionals do not possess, while vendors are often not accountable for mistakes during implementation.

In summary, there is a significant gap between the political vision and the objectives for data sharing (such as data spaces, a fair data economy, and enhanced innovation and research) and the technical ability to implement these goals in a privacy-preserving manner, especially for microdata within big-data contexts.

Most fundamentally, the current over-optimistic portrayal of PETs for data sharing overlooks that, in most cases, one is not facing technical challenges but rather policy ones. If preserving data privacy is to be taken seriously, there needs to be a paradigm shift in the way practitioners and policymakers currently think about facilitating (micro)data sharing in privacy-sensitive areas: Away from technology-centric recommendations towards purpose-driven approaches. Instead of relying solely on technologies to resolve it, future policies and guidelines need to acknowledge the fundamental trade-off between privacy and utility inherent to any data use.

This policy paper puts forward eight recommendations for policymakers and data practitioners to implement this shift. These recommendations support a realistic approach to privacy-preserving data sharing and emphasize, in particular, data purpose-driven decision-making, risk-based frameworks, and the evaluation of technical solutions in alignment with privacy utility trade-offs.

Table of contents

Executive summary	02
1. Introduction	04
1.1 Balancing innovation and privacy: ambitions and challenges for data sharing	04
1.2 The (broken) promise of privacy-enhancing technologies for data sharing	06
1.3 Synthetic data as a case study	06
1.4 The need for realistic policy approaches to privacy-preserving data sharing	07
2. The fundamental limits of privacy-preserving data sharing	09
2.1 Defining utility: the promised benefits of data sharing	09
2.2 Defining privacy: the potential harms of data sharing	10
2.3 The fundamental trade-off of privacy-preserving data sharing	11
2.4 The overly optimistic portrayal of PETs	11
3. Recommending privacy technology – a difficult practice	14
3.1 Synthetic data: a case study on the limits of privacy-preserving data sharing	14
3.2 Can synthetic data meet utility demands?	15
3.3 Market asymmetries in the data supply chain	16
4. The risks of maintaining a technology-centric approach to data sharing	18
5. Policy recommendations	20
6. Conclusion	23
References	24

1. Introduction

1.1 Balancing innovation and privacy: ambitions and challenges for data sharing

*Data sharing*¹ has become a key priority on the European Union's political agenda. By spanning aspects of data availability, access, control, exchange and pooling, data sharing holds the potential to drive innovation, improve public services, and democratize access to data. It is already of critical importance in the sectors of finance (FCA, 2024), medical research (TEHDAS, 2023), and manufacturing (Frost and Sullivan, 2019; IBM, 2020). Access to vast amounts of data has further become paramount with the rise of generative artificial intelligence (AI) and the EU's ambitious AI strategy (European Union, 2025).

The EU's 2020 *European Strategy for Data* aims to create a legal framework that promotes trust and fairness in the data economy and to establish a single market for data to ensure Europe's global competitiveness and data sovereignty (European Commission, 2020). Importantly, this includes establishing sectoral data spaces. Data spaces are conceptualized as "sovereign, trustworthy and interoperable data-sharing environments where data can flow within and across sectors, in full respect of European ambitions, rules, and values" (Farrell et al., 2023, p. 7). This vision of data spaces encompasses personal and non-personal data, sensitive business data, and high-quality industrial data (European Commission, 2020). This vision also addresses the concentration of power in private global companies that control data infrastructures, as this power limits data availability for other market players, especially European companies (Ada Lovelace Institute, 2022; Kolbe-Guyot and Finger, 2025).

The strategy, along with several successive cross-sectoral legislative instruments – such as the 2022

Data Governance Act (DGA, European Union, 2022a) and Digital Markets Act (DMA, European Union, 2022b) as well as the 2023 Data Act (European Union, 2023) and Implementing Act on High-Value Datasets (European Commission, 2023) – envisions a single market for facilitating data sharing and for creating a digital ecosystem of data intermediaries (such as data marketplaces). As an alternative to monopolistic platforms, such an ecosystem could take the form of data-sharing pools and data cooperatives for big-data analytics and machine learning (European Commission, 2020). The DGA and the Implementing Act on High-Value Datasets already regulate the publishing and re-use of publicly held protected data by mandating data sharing by public bodies and private companies that hold high-value datasets. Most importantly, the EU Data Act stipulates the setting up of (now) fourteen sectoral data spaces that are meant to facilitate access, interoperability, and the legitimate reuse of data (European Commission, 2023). Even beyond the EU, the development of data spaces has garnered increasing political interest, as is the case in Switzerland (Swiss Federal Council, 2022; Kolbe-Guyot and Finger, 2025).

One of the central purposes of data spaces outlined in the *European Strategy for Data* is to make high-value datasets available for re-use and secondary purposes.

Today, a significant vendor market for data-sharing platforms and technologies has emerged (see Farrall et al., 2022; VentureRadar, 2022), ready to serve the future need for data sharing through these data spaces.

¹ The terms 'data sharing' and 'data publishing' are often used synonymously; however, for this paper, 'data sharing' specifically refers to the transfer of micro-datasets from original owners to third parties. This transfer can occur via legal agreements or controlled public release. Our focus is on situations where data is explicitly published and expected to leave the original owner for potential pooling elsewhere, rather than on federated access methods.

One of the central purposes of data spaces outlined in the *European Strategy for Data* is to make high-value datasets available for re-use and secondary purposes (European Commission, 2020).² In this context, the sharing of *microdata*, i.e., datasets that capture information about individuals (also referred to as record-level or individual-level data) plays a significant role due to two main characteristics. First, in contrast to aggregate macro-statistics, microdata contains fine-grained statistical patterns that are needed for certain use cases: Financial fraud and medical-anomaly detection are two frequently mentioned examples of data use enabled through wider data-sharing (FCA, 2024; TEHDAS, 2023). To fulfil this need, it is essential that the data shared through data spaces must “retain the information richness” (Hradec et al., 2022, p. 2) of the original datasets.

The second characteristic is that microdata is flexible in its use. A single dataset can serve as the basis for a variety of analysis tasks, including “the basis for more advanced research queries” (TEHDAS, 2023, p. 25). This means that the shared data should not only be useful to run well-defined analytical computations, but that it should also preserve the original data’s utility for novel types of analysis not known at the time of the data sharing. Indeed, many of the envisioned benefits of data spaces appear to depend on the broader sharing of microdata, sometimes also referred to as data publishing, rather than merely providing access to aggregate macro-statistics or computational interfaces designed for specific statistical analyses which could also be implemented through privacy-preserving cryptography.

Data-driven innovation is seen as instrumental in the digital transformation of European society and organization. But, it should not come at the expense of privacy rights.

One of the primary challenges in the establishment of data spaces that also contain microdata is the current limitation on data sharing. This arises from *privacy*³ concerns and concomitant legally restricted access to microdata (such as with the General Data Protection Regulation – GDPR). In addition to more

traditional microdata, such as census data, survey data, and data from clinical trial results, much of behavioral data is currently generated through human interaction via digital devices, such as smartphones and fitness trackers (Gadotti et al., 2024, p. 1). Mobile-phone data, social-media data, and even consumer data from cars and cyber-physical systems, such as smart meters and internet-connected medical devices, constitute comprehensive, *high-dimensional datasets*⁴ that are invaluable for a wide range of applications and analysis.

However, due to the detailed information they contain about individuals’ behavior, these datasets also pose particular risks to individual privacy (Narayanan and Shmatikov, 2008). As has been repeatedly shown in practice, today’s information-rich datasets often allow for the inference of highly detailed patterns about our personal lives, such as our daily whereabouts, sexual orientation, and religious habits (Sweeney, 2000; Narayanan and Shmatikov, 2010; De Montjoye et al., 2013). Without appropriate safeguards in place, the widespread sharing of such data could lead to significant individual and societal harms, such as economic and financial loss, psychological distress, reputational damage, discrimination, and physical threats (Solove, 2024; Citron and Solove, 2021).

Therefore, GDPR, along with various other privacy legislations⁵, imposes strict controls on the sharing of personal and sensitive data. This leads to a complex and often burdensome process for entities seeking access to the data they require (Jordon et al., 2022, p. 8). At the same time, in the context of the current vision of European data spaces, data-driven innovation is seen as instrumental in the digital transformation of European society and organization (Granell et al., 2022). As a result, a potential tension emerges between the promotion of data access and the protection of privacy rights: Data-driven innovation is vital but should not come at the expense of privacy rights (Hurtado Ramírez et al., 2023, p. 4). Therefore, data sharing that is privacy-preserving has been elevated to an important functional requirement of future data-space architectures (Farrell et al., 2023, p. 23), public-sector data release (European Commission, 2023), and the decentralized data-sharing economy at large (Farrell et al., 2023, p. 24).

² ‘Re-purposing of data’ or ‘secondary use of data’ refers to the analysis of existing data collected by other individuals or entities for a new research purpose; this means that data is being adapted to serve new objectives or applications beyond its initial scope.

³ Although there exist different understandings of privacy, here we understand ‘privacy’ as referring to confidentiality.

⁴ ‘High-dimensional datasets’ are datasets with vast amounts of information (i.e., many attributes).

⁵ See for example, Swiss Federal Assembly (2020).

1.2 The (broken) promise of privacy-enhancing technologies for data sharing

Amidst the increasing pressures to facilitate privacy-preserving data sharing, privacy-enhancing technologies (PETs) have been promoted by advisory bodies, policy consultants, and solution vendors to be key technologies to resolve the tension between exploiting the benefits of broadened data use and protecting individuals' privacy (Hurtado Ramírez et al., 2023; Farrell et al., 2023). PETs for data sharing are frequently portrayed as technological “silver bullets” that can enable the free sharing of data for re-use and secondary purposes, as envisioned by the *European Data Strategy*, while effectively mitigating any privacy risks. The problem with this vision is that it is often at odds with the realities of the current and future capabilities of these technologies: Privacy-preserving data-sharing technologies are subject to certain fundamental trade-offs between preserving a dataset's value for statistical analysis and protecting individuals' private information.⁶

A data-sharing technology offering high utility cannot at the same time also ensure strong privacy protection. Either the shared data is flexible with a higher risk of private information inference, or it minimizes privacy risks by limiting its use. This trade-off is intrinsic to all privacy-preserving data sharing, not just specific technology.

At the heart of this trade-off lies a fundamental dilemma that no PET can overcome: In order to derive value from data sharing, it is necessary that the statistical properties of the shared data closely match those of the original data. However, any information that must be preserved for utility reasons could also be misused by *privacy adversaries*⁷ to infer sensitive information about individuals and inflict harm. Conversely, the stricter the privacy guarantees are, the more the statistical properties of the shared data deviate from the original data hence decrease flexibility in utility. Yet, this trade-off often runs counter to the promised value of open data spaces that require access to “*general purpose, high utility*”

data, i.e., datasets that contain information useful for a variety of analysis tasks that might not be known or not well-defined at the time of the data sharing.

As a result, any data-sharing technology that provides high utility for the envisioned use cases (such as secondary use of data research and innovation) cannot, at the same time, provide strong protections against privacy attacks. The shared data can either be highly flexible in its use, which means a higher risk of private information inference, or effectively mitigate potential privacy risks, which involves strictly limiting its use for a particular purpose. This trade-off is not limited to any specific technology; rather, it is an intrinsic limitation of privacy-preserving data sharing as a whole.

Current policies and data-sharing strategies, however, fail to adequately acknowledge this reality. The widespread assumption is that PETs either already provide or will eventually (after some more research) provide solutions for data sharing, without compromising on utility or privacy. This is illustrated by the way the well-known shortcomings of earlier anonymization techniques are often dismissed as specific flaws of those methods, instead of being acknowledged as a result of the fundamental trade-off between privacy and utility. Instead of recognizing that significant privacy risks are an inevitable consequence of maintaining information utility, the prevailing discourse optimistically assumes that new or alternative data-sharing technologies will eventually resolve this inherent challenge (Hradec et al., 2022; Hurtado Ramírez et al., 2023; Jordon et al., 2022).

1.3 Synthetic data as a case study

The issue of balancing privacy and data utility is not specific to any single technology; it is an inherent challenge across all data-sharing methods. We use *synthetic data generation* (SDG) to illustrate this issue in greater detail.

Often marketed as an innovative solution that overcomes the shortcomings of traditional anonymization, synthetic data is purported to deliver the desired data utility while safeguarding privacy (Bellovin, Duttav and Reitering, 2019). SDG refers to a set of techniques

⁶ For instance, the fundamental law of information recovery, see Dinur and Nissim (2003) and Dwork and Roth (2014).

⁷ ‘Privacy adversaries’ is the commonly used technical term to refer to unknown individuals or entities who attempt to misuse or steal data with malicious intent, often with the goal of inferring some information about individuals or their attributes from the information contained in the dataset. See also [table 2](#).

designed to produce “artificial” data derived from real datasets.⁸ Synthetic datasets are datasets that are created from statistical models fitted on real data and that maintain (some part of) the original data's statistical traits but do not permit a 1:1 link between real and “artificial” data points (Drechsler, 2011). The aim is to create and share microdata, thus enabling analysts to conduct analyses with the same methods and processes they would apply to actual data while ensuring privacy protection (Gadotti et al., 2024, p. 7).

Synthetic data is often seen as a solution for the balance of privacy and utility, but it faces challenges in preventing privacy vulnerabilities.

Synthetic data is appealing because it is argued to preserve privacy and to overcome the limitations of traditional anonymization techniques that have been demonstrated to be vulnerable to re-identification attacks (Narayanan and Shmatikov, 2008).⁹ By promising true privacy, at almost no cost to utility, synthetic data could also be viewed as GDPR conform (Meeus et al., 2024, p. 381). As synthetic data consists of “fully artificial” records, some practitioners and advisory bodies have argued that it is inherently private and safe to share (Bellovin et al., 2019; Hradec et al., 2022; IEEE, 2023); some even promote it as a direct replacement for the original data (Hradec et al., 2022). As a result, influential actors from the sectors of research, policy, industry, and civil society have championed synthetic data as a key solution for data sharing in privacy-sensitive areas; they recommend it to policymakers and data owners (as prospective sharers) for realizing wider access to sensitive datasets for secondary data uses, such as research and innovation.¹⁰

Although this is certainly an appealing proposition that aligns well with the current hopes attached to the future European-data economy, synthetic data is far from the “holy grail” of privacy-preserving data sharing: Empirical tests have shown that synthetic data can *leak information*, i.e., unintentionally expose sensitive or private information from the original data-

set, thus making it vulnerable to *privacy attacks* (Lu, Wang and Yu, 2019; Yale et al., 2019; Stadler, Oprisanu and Troncoso, 2022; Annamalai, Gadotti and Rocher, 2024). These findings highlight a crucial aspect of the privacy-utility trade-off: The unintended leakage of private information is inherent to any high-utility dataset, whether anonymized or synthetic. Therefore, the privacy-utility dilemma cannot be resolved solely through technological advancements.

1.4 The need for realistic policy approaches to privacy-preserving data sharing

Current policies, guidelines, and recommendations for data-sharing technologies often fail to acknowledge the fundamental limitations of PETs. Instead, they focus on discussing the various pros and cons of specific technologies, hence implicitly or explicitly assuming that future research, standardization, and proper implementation will resolve the tension between data use and privacy risk (see Hurtado Ramírez et al., 2023). This paper contends that although PETs, including some SDG methods, can offer substantial privacy protections, they ultimately fall short of ensuring both privacy and high utility for a broad range of data-analysis tasks. No privacy technology can perfectly preserve the utility of the original microdata for “all-purpose” analysis and simultaneously guarantee the protection of sensitive personal information, without any compromise.

Yet, the discussions surrounding data sharing are characterized by a highly problematic trend of creating blanket promises about technologies that purport to deliver all-purpose privacy-preserving data-sharing solutions, without sufficiently acknowledging necessary trade-offs. Such unrealistic promises are not only misleading but also contribute to an erroneous belief that technology can effortlessly overcome fundamental limitations in data sharing and publishing. It is important to reconcile these overly optimistic notions with the practical realities of PETs at large and, in particular, with synthetic data: They can either achieve strong privacy or fulfil the intended all-pur-

⁸ Synthetic data comes in two main forms: partially synthetic and fully synthetic. Partially synthetic data replaces the values of specific sensitive attributes with values generated by a machine-learning model trained on supposedly non-sensitive attributes, thus creating “artificially sensitive” attributes. Whereas, fully synthetic data involves sampling records entirely from aggregate data through a data generation model, thus ensuring that the original data is never shared directly, only used for training the model. See Gadotti et al. (2024, p. 11).

⁹ A ‘re-identification attack’ is the process of inferring private, sensitive information about individuals from a seemingly anonymized dataset. See also Gadotti et al. (2024) for further explanations and definitions.

¹⁰ See, e.g. the publications of IAPP (2010), Bellovin et al. (2019), Hradec et al. (2022), TEHDAS (2023), Farrell (2023), FCA (2024), Joint Research Center (2024), MostlyAI (2025), and SynthIA (2025).

pose analysis function, but never both at the same time.

No anonymization technology can achieve, without compromise, both data utility and privacy. Yet, discussions often make overblown promises about all-purpose solutions without addressing inherent trade-offs.

It is essential for policymakers to acquire a comprehensive understanding of both the capabilities and limitations of technical solutions to privacy-preserving data sharing, including synthetic data. Informed policy discussions and law making hinge on such knowledge and are necessary to avoid the pitfall of crafting policies or technical endorsements, based on unrealistic expectations. At the moment, a significant gap between political vision and objectives (e.g., data spaces, a fair data economy, improved innovation and research) and the technical ability to provide them exists. Although we agree that there is a clear need for privacy-preserving data-sharing technologies, expectations about their capabilities need to be more realistic.

Policymakers should recognize that any technology that provides strong formal privacy guarantees will not be able to simultaneously produce the desired “all-purpose” data for secondary use and research. There are no technical “silver-bullet solutions” to the privacy-utility problem, and they might never exist. Instead, policymakers should offer procedural guidance on managing the risks inherent to balancing privacy protection and flexible data utility in order to enable data practitioners to make informed judgments.

Although we agree that there is a clear need for privacy-preserving data-sharing technologies, expectations about their capabilities need to be more realistic.

In section 2, we will define data utility and privacy risk in data sharing and explain the inherent trade-off between the two, detailing the reasons an overreliance on technical solutions for solving the privacy-utility dilemma is problematic. In section 3, using synthetic data as an emblematic example, we explore the issues with recommendations for specific privacy technologies within data-sharing contexts, and we emphasize the need for more informed

approaches. In section 4, we critique the reliance on technology-centric data-sharing frameworks and stress the need for procedural methods that align with data-sharing purposes and that effectively address privacy concerns. Concluding the paper in section 5, we propose recommendations for policymakers and data practitioners to shift from a technology-centric to a more policy-focused approach to data sharing.

2. The fundamental limits of privacy-preserving data sharing

2.1 Defining utility: the promised benefits of data sharing

The promise of privacy-preserving data-sharing technologies is for data owners to be able to share data that, on the one hand, preserves its analytical value but, on the other hand, protects the privacy of individual data subjects. But what defines the analytical value of a dataset? What defines its utility? Data-sharing scenarios and discussions about the use of PETs are rarely developed with these questions in mind and in sufficient depth.

The utility of data can be assessed in different ways, including through statistical measures of error or differences in performance across data-analysis tasks. Yet, in practice, it is the specific objectives underlying the analysis tasks that ultimately determine a dataset's utility: Whereas the same dataset could be highly useful for one task, it could be rather ineffective for another (Gadotti et al., 2024, p. 3). Utility also varies across different use cases and

intended audience groups (e.g., an internal analyst (targeted) or the public/certified researchers (broad)). Taken together, the utility of data is determined by the *intended purpose of the data sharing*, i.e., (1) the breadth and accessibility of use and (2) the specificity and predictability of analytical tasks to be performed. To better understand the utility and varying purposes of data sharing, we distinguish in table 1 between three intended purpose categories: *all-purpose use*, *unknown statistical analysis*, and *known statistical analysis*.

The value of data sharing, whether it be for use cases of economic, research, or public interest, depends on the anticipated purpose and the subsequent flexibility of analytical tasks it can support. Reviewing many of the documents listed in section 1.1, we note that the main cited purposes of data sharing fall into the first two categories: all-purpose data sharing (e.g., open data) and the secondary use of data for research and innovation. The third category of purpose is less frequent.

09

Table 1 | Categories of data-sharing purposes

Category	Description	Example
All-purpose use	A broad purpose category characterized by data accessibility and utility to a wide range of users, including non-tech-savvy individuals. The goal is to democratize data, thus enabling exploration and analysis by virtually anyone.	Open data initiatives such as the European Data Portal
Unknown statistical analysis	A narrower purpose category targeting a specific audience to uncover trends, correlations, or anomalies not previously identified. The value in the shared data lies in discovering new, unknown patterns, even if the exact data value and quality are initially difficult to assess.	Secondary use of health records to discover new disease patterns or treatment outcomes
Known statistical analysis	This scenario involves a clearly defined value and purpose for sharing data, with specific analytical tasks established beforehand. It allows for more accurate utility evaluation due to pre-existing criteria for analysis.	Financial fraud detection

2.2 Defining privacy: the potential harms of data sharing

In the same way that the purpose of the data sharing defines its utility requirements, each data-sharing use case also comes with specific *privacy risks* that the use of PETs is intended to mitigate.

The datasets at the center of current data-sharing initiatives contain detailed sensitive information about individuals. From electronic health records that detail a patient's entire medical history to credit card metadata that reveal an individual's location and behavioral preferences, today's information-rich datasets carry a large potential not only for valid statistical analysis but also for inferring highly specific patterns about people's personal lives. The inference of such personal, sensitive information by ill-intentioned actors, a.k.a. *privacy adversaries*, opens the way to concrete privacy harms such as mass surveillance, increased risks of blackmail, harassment, and identity theft (Wood et al., 2022; Ohm, 2010).

The purpose of PETs for data sharing is to produce datasets that protect individuals from such inferences and their resulting harms. *The use of PETs in data-sharing scenarios always means that sharing the original data would entail certain privacy risks, and that these risks have been identified.* The potential privacy risks of sharing individual-level microdata typically include, but are not limited to, membership

inference (or table linkage), re-identification (or unique record linkage), and attribute disclosure (Sweeney, Loewenfeldt and Perry, 2018; Article 29 Data Protection Working Party, 2014). Each privacy risk is defined by an *adversary model*. The adversary model describes the inference goals and capabilities of the adversary, i.e., the (potentially harmful) information an adversary is trying to gain from the shared data and the means they might use to get to their goals.

Together with data-utility requirements (see section 2.1), the privacy risks of sharing a particular dataset and the underlying adversary model determine which PET would be most appropriate to use: The PET in question must effectively mitigate the privacy risks of sharing the original data. To evaluate whether a PET achieves this, it is necessary to first define the privacy concerns of the data-sharing scenario, then to assess the risk of sharing the original data, and finally to evaluate the extent to which a PET reduces these risks.

Following this process could avoid two major pitfalls in privacy-preserving data sharing. First, the use of PETs and their associated utility loss can be avoided, in cases where already sharing the original data poses little to no privacy risks. Second, the sharing of data through PETs that do not sufficiently protect against the relevant risks and leave individuals vulnerable to re-identification attacks can be prevented (Altman, Cohen and Nissim, 2024).

Table 2 | Definitions of different data privacy and security terms

Aspect	Description
Privacy risks	Potential threats to individuals' privacy arising from the collection, handling, sharing, or misuse of personal data.
Privacy harms	Negative effects from misuse or exposure of personal data, manifesting in various forms such as physical, economic, reputational, discriminatory, relational, psychological, and autonomous impacts. These harms often begin as subtle, minor incidents that accumulate across various actors over time. Their full extent is not immediately apparent, increasing the risk of future repercussions and highlighting their broader societal implications (Citron and Solove, 2021).
Privacy adversaries	Unknown individuals or entities attempting to misuse or steal data with malicious intent. Their goal is to infer information about individuals or their attributes from datasets.
Privacy attacks	Attempts by adversaries to gain unauthorized access to data, identify or re-identify data points, track individuals, analyze their attributes, and combine data with other sources in potentially harmful ways.
Re-identification attacks	Acts of deducing private information from anonymized data.

2.3 The fundamental trade-off of privacy-preserving data sharing

The promise of PETs is to enable the free sharing of data, across a large spectrum of use cases, with high utility and strong privacy-protections (Joint Research Center, 2024; TEHDAS, 2023). However, no PET could ever fulfil this promise *for the use cases that require high flexibility in data utility, i.e., all-purpose data uses such as research and innovation*.

At the core of this problem lies a dilemma that no technological solution can mitigate: Any information retained for valid statistical analysis could also be misused to inflict harm by privacy adversaries. This fundamental tension between the utility and privacy risks of a dataset is a well-established result within the technical privacy research community (Dinur and Nissim, 2003; Dwork et al., 2009; Ullman and Vadhan, 2011). However, the implications of this trade-off for the practical applications of PETs are rarely acknowledged: It is impossible for any PET to produce datasets that retain the original data's *general purpose* utility while effectively mitigating against privacy attacks (Naryanan and Shmatikov, 2008; Narayanan and Shmatikov, 2010; Stadler, Oprisano and Troncoso, 2022).

If a dataset preserves all of the statistical signals necessary to serve a large varied user group or to enable the discovery of novel, unknown patterns (incl. anomalies), it also always retains enough information to draw unintended inferences that infringe on individuals' privacy. Whereas if a dataset effectively mitigates the privacy risks of sharing the original data, this means that some of the original data's information must have been removed to hide it from privacy adversaries. As a result, the shared data can no longer be considered "all-purpose": The information that has been removed might include statistical properties that are required for certain data uses such as advanced research queries.

Hence, we must recognize that every data use case comes with some inevitable privacy risks. These inherent risks are defined by the intended purpose of the data sharing. Any information that we must preserve to fulfil the utility requirements of the data sharing could also be misused by privacy adversaries and lead to harmful inferences.

2.4 The overly optimistic portrayal of PETs

Policy documents, guidelines, and strategies, however, often do not acknowledge the inherent limitations of (open) data sharing for the intended purposes. Instead, they tend to portray the trade-off between utility and privacy as a technology-specific issue. In particular, the inability to share "high-utility, general-purpose" datasets while preserving privacy is often blamed on the shortcomings of traditional anonymization¹¹ techniques such as pseudonymization¹² or de-identification¹³. With regard to these technologies, the utility-privacy trade-off has been recognized as a considerable issue.

It is widely acknowledged that creating anonymized data *that remains useful for research and other secondary data uses* inevitably involves significant privacy risks: Pseudonymization and de-identification cannot prevent privacy attacks by adversaries with access to extensive auxiliary information and fail to protect privacy for modern high-dimensional data (Gadotti et al., 2024, p. 3–6). The failures of traditional anonymization techniques have been repeatedly demonstrated by high-profile cases involving, for instance, data from AOL and Netflix.¹⁴ It is particularly the big data context that has made the privacy-utility trade-off apparent: Achieving meaningful privacy in high-dimensional data often requires increasing generalization and perturbation to levels that significantly undermine its utility (Soria-Comas and Domingo-Ferrer, 2016).

11

¹¹ 'Anonymization' involves processing personal data in such a way as to create information from which individuals cannot be identified. Thus, anonymized data does not allow tracing back information to a specific individual and hence should be resistant to privacy attacks that seek to (re-) identify individuals or extract sensitive information about them (Gadotti et al., 2024, p. 1).

¹² 'Pseudonymization' refers to the removing or replacing of direct identifiers, i.e., personally identifying information such as names, phone numbers, or insurance numbers with random pseudonyms (Gadotti et al., 2024, pp. 3–4).

¹³ 'De-identification' refers to the modification of individual information through a set of specific techniques, such as k-anonymity, that generalize or perturb certain quasi-identifiers (personally identifying information such as date of birth, gender, or ZIP code) to make individual records become to a certain degree indistinguishable from others regarding these attributes (Gadotti et al., 2024, pp. 5–6).

¹⁴ Researchers have demonstrated that individuals' online search queries and movie viewing histories can be used as 'personally identifying' attributes to re-identify supposedly anonymous users. This is largely due to the issue of sparsity in high-dimensional data: the vast number of attributes makes each combination in a given record nearly unique, and hence easy to re-identify. See Narayanan and Shmatikov (2008).

There is a reluctance to recognize that these anonymization failures stem from an inherent trade-off between utility and privacy, not just current technological shortcomings.

Many privacy researchers and practitioners have drawn two main lessons from the historical failures of traditional anonymization: (1) Privacy approaches are needed that do not rely on defining and removing “personally identifying information (PII)” among a dataset’s attributes. And (2) the high dimensionality and sparsity of modern datasets pose significant challenges for privacy. These lessons have steered public and policy discourse towards flawed recommendations for technologies, such as SDG, as they are believed to offer a solution to both problems. There is a reluctance to recognize the core issue of the problem: Previous failures to protect privacy stem from the inherent utility-privacy trade-off of data sharing and not only from technological shortcomings. Instead, there persists a belief, about emerging PETs and “novel anonymization techniques”, that they promise high utility and minimal privacy risks.

The ongoing policy discussions surrounding the “right” technologies for privacy-preserving data sharing often overlook the crucial aspect of the *desired utility of shared data*. Many current proposals advocate for enhanced privacy through PETs, such as differentially private¹⁵ aggregation. Although these methods provide robust privacy protections, they significantly restrict the data’s usability for certain applications. It is therefore important to recognize that differential privacy (which is a data-*processing* mechanism) might not be the most suitable model for general-purpose data sharing.

Similarly, the lack of utility considerations has also led to the proposal of PETs, such as federated learning¹⁶ or secure multi-party computation¹⁷, that strictly limit the use of the data to a concrete analysis task, i.e.,

these technologies are tailored to carry out particular functions or analyses and do not allow for broader or different uses of the data beyond the intended task. They also do not entail the direct sharing of data, or a discreet publishing of data. Rather, they securely connect the data holders to enable them to run analyses, without the data leaving its owner. Although this also offers certain privacy protections, it stands at odds with the European vision of data spaces, where data pools are built for big-data analytics and machine learning (European Commission 2020, 2022). Consequently, these technologies do not, by design, allow for broad data-sharing (purpose category one, see [table 1](#)) and cannot fulfil the desired purposes of more versatile or unrestricted data usage, where the precise analysis tasks might not be fully known at the time of data publishing (purpose category two).

Therefore, what makes realizing the current vision of data sharing in privacy-sensitive contexts so challenging? The deciding factor in determining privacy risks is the amount and type of information the shared data retains about the original data. Any information that is preserved to fulfil utility requirements can also be repurposed and misused by adversaries to infer private information. This means that the more information we need to share in order *to realize the value of the data*, the higher the *inherent privacy risk of the data sharing is*. This underscores a fundamental reality: the privacy-utility trade-off cannot be “out-engineered.” The core challenge of privacy-preserving data sharing comes from the need to retain identifiable information for utility purposes, not only the data’s dimensionality.

The more information we need to share to realize the value of the data, the higher the inherent privacy risk of the data sharing is. This underscores a fundamental reality: the privacy-utility trade-off cannot be “out-engineered.”

¹⁵ ‘Differential privacy’ is defined as privacy that facilitates the public sharing of information about a dataset while enabling the monitoring and limiting of potential information leakage. In practice, this is achieved by introducing randomness, or “noise,” to the processed data, which makes it difficult to identify specific individuals, yet still enables the derivation of meaningful aggregate insights from the data. Differentially private mechanisms offer a mathematical guarantee that the information of a particular individual cannot be identified from the disclosed data (Wood et al., 2018).

¹⁶ ‘Federated learning’ is a machine-learning technique that trains a machine-learning algorithm across multiple decentralized devices or servers that hold local data samples, without exchanging them. To train the global algorithm, different parties first train local models on their datasets and then exchange the parameters of these models with each other. Thus, only model parameters, but not the data itself, are revealed (Yin, Zhu and Hu, 2021).

¹⁷ ‘Secure multi-party computation’ is a cryptography subfield that enables multiple parties to compute a function jointly over their data, without directly sharing it with others or a trusted third-party. It is used when parties with private data need to compute a function by using all the data, but they also seek to maintain their data’s privacy. It also sidesteps potential problems associated with merely pooling the data with a trusted third party, such as inadvertently creating a single point of failure, and it eliminates the necessity for a universally trusted third party to ensure privacy and accuracy of results (Cramer, Damgard and Nielsen, 2015).

In summary, erroneous lessons were drawn from the shortcomings of traditional anonymization. Rather than acknowledging the inherent privacy risks in sharing high-utility general-purpose datasets, researchers and policymakers place their faith in novel PETs for data sharing. However, as has been argued here, the belief that these technologies can simultaneously provide robust privacy protection and high utility for flexible data analysis is overly optimistic. Genuine privacy, which aims to prevent harm rather than merely obscure personally identifying information, is attainable only if we can clearly define the intended purpose of the data sharing and if this purpose enables us to restrict the amount of sensitive information that must be shared. This presents a fundamental limitation that no PET can overcome.

3. Recommending privacy technology – a difficult practice

3.1 Synthetic data: a case study on the limits of privacy-preserving data sharing

Regardless of the fundamental limitations, there is a persistent trend of searching for and recommending novel technologies as an alternative to traditional anonymization; each time promising to resolve the data-sharing challenge and to ensure both high utility and high privacy. Actors of the public, research, and private sectors frequently advocate for specific data-privacy technologies.¹⁸ However, as some of the problematic assumptions underlying these technologies reveal, such recommendations are a tricky practice and often difficult to reconcile with the empirical evidence of delivering the promised solution to the privacy-utility trade-off. *Synthetic data*, which has recently attracted much interest from data practitioners (Jordon et al., 2022), is probably the most notorious example of this. In this context, we can identify three main perspectives on the role of synthetic data in privacy-preserving data sharing; we summarize them in table 3.

The *optimistic view* on synthetic data presents a strong belief in its potential as a privacy-preserving data-sharing technology. Its advocates, mainly composed of synthetic-data vendors¹⁹ but also some research²⁰ and scientific advisory bodies²¹, promote synthetic data as an ideal solution for balancing privacy and utility in large-scale data sharing and in ambitious applications such as data-driven policy-making and digital twins. The optimistic view on synthetic data emphasizes the benefits of data shar-

ing while minimizing its technical limitations as a data sharing technology. Issues of unintended information leakage and privacy failures are often attributed only to issues of implementation.²²

Table 3 | Perspectives on SDG

Perspective	Description	Advocates
Optimistic	Strong belief in synthetic data's potential as a privacy-preserving data-sharing technology, focus on benefits for business, innovation, and public policy.	Mostly composed of synthetic data vendors and advisory bodies
Pragmatic	Recognizes enduring technical challenges with synthetic data but maintain an optimistic outlook about mitigating these issues.	Composed of industry entities and public research organizations
Skeptical	Argues that synthetic data cannot fully solve the privacy-utility trade-off, only minimize it, due to unavoidable inherent privacy risks.	Mainly academic researchers

The *pragmatic view* represents a more nuanced perspective that explicitly recognizes the enduring privacy challenges associated with synthetic-data technologies while maintaining an optimistic outlook. Data-driven industry entities²³ and public research

¹⁸ See, e.g., IAPP (2010), Bellovin, Dutta and Reiter (2019), Jordon et al. (2022), Hradec et al. (2022), TEHDAS (2023), IEEE (2023), Farrell (2023), FCA (2024), Joint Research Center (2024), MostlyAI (2025), and SynthIA (2025).

¹⁹ For example, MostlyAI (2025).

²⁰ For example, the van der Schaar lab at the University of Cambridge [\[2\]](#), or the Electronic Health Information lab at the University of Ottawa [\[3\]](#).

²¹ For example, Hradec et al. (2022), Farrell et al. (2023) and IEEE (2023).

²² For example, Hradec et al. (2022, p. 58) claim that “[...] synthetic data are free of privacy issues when well designed and quality checked.” However, in-depth discussions about the actual technical implementation (and concomitant complex difficulties) are mainly sidelined.

²³ For example, FCA (2023, 2024).

organizations²⁴ that take a pragmatic view on synthetic data acknowledge issues such as information leakage and vulnerability to privacy attacks. They believe these challenges can be mitigated through continued research and proper implementation, including the careful selection of appropriate use-case scenarios. The pragmatic view attributes difficulties in generating synthetic data, which is both privacy-preserving *and* high-utility, to the lack of technological maturity and the need for standardization and best practices (see FCA 2023, 2024; Jordon et al., 2022).

The pragmatic view advocates for the use of synthetic data in specific applications, emphasizing the importance of selecting technologies tailored to individual scenarios. It thus implicitly recognizes that synthetic data might not fully achieve broad data-sharing objectives. Unlike the optimistic view, a pragmatic view on synthetic data acknowledges privacy risks in data sharing and calls for risk assessments based on inference capabilities. Consequently, from a pragmatic point of view, differentially private synthetic data is often recommended as the best solution for balancing privacy and utility. However, this view does not sufficiently acknowledge that such techniques can fall short of meeting practical utility demands, such as the need for general-purpose (micro)data for certain real-world applications (European Commission, 2020, 2022).

The *skeptical view*, mainly taken by academic researchers, contends that synthetic data, like any other PET, cannot fully solve the problem of privacy-preserving data sharing due to the fundamental privacy-utility trade-off.²⁵ It stresses that (micro) datasets, which are useful for the envisioned data use-cases, always risk privacy breaches due to the detailed information that must be retained; a trade-off that cannot be overcome with more research or better standardization. If we seek high utility for research or secondary use of data, achieving privacy becomes impossible (Stadler, Oprisanu and Troncoso, 2022; Annamalai et al., 2024). Instead of relying on purely technical solutions in data sharing, the skeptical view strongly advocates accepting the inherent risks of specific data use cases tied to their intended purpose and hence employing procedural policies that guide prudent decision-making in data-sharing contexts.

In summary, many proponents believe that the utility-privacy trade-off in data sharing can ultimately be resolved through technical solutions. They tend to emphasize the pros and cons of privacy technologies, assuming future research will address current issues and often downplaying critical privacy limitations. This stance is unwarranted, however, given the inherent challenges and limitations of current technologies, as well as empirical evidence questioning their effectiveness in fully reconciling privacy with data utility for the desired data-uses.

3.2 Can synthetic data meet utility demands?

Most proponents of SDG offer only insufficient evidence for their claims that synthetic data can deliver good utility for data uses such as advanced research queries. Empirical evaluations of these cases report mixed results. For example, Annamalai et al. (2024) find that whereas basic synthetic data generation does not balance privacy and utility well, some new methods, such as differentially private synthetic data, offer better results but only when using parameters that do not strongly assure robust privacy. Evaluating synthetic data for training machine-learning models and for outlier analyses, Stadler, Oprisanu and Troncoso (2022) find that the trade-off between privacy and utility offered by synthetic data is not superior to traditional anonymization techniques, including generalization and k-anonymity. Lastly, although Gadotti et al. (2024, p. 12) acknowledge the usefulness of synthetic data for testing new systems and exploratory analyses, they stress that its accuracy is highly dependent on the specific application, hence necessitating validation against the original data to confirm any findings. This is particularly troubling, as a lack of accuracy can, for instance, adversely affect the quality of public decision-making and the research based on that data (Schneider, 2021).

The usefulness of synthetic data for achieving the current all-purpose utility vision of data sharing is further complicated by the need for case-specific approaches and limitations of linking synthetic data. First, several advisory bodies have recommended

²⁴ For example, Jordon et al. (2022).

²⁵ For example, Stadler, Oprisanu and Troncoso (2022), Gadotti et al. (2024), Meeus et al. (2023), Yao et al. (2025), and Ganev, Oprisanu and De Cristofaro (2022).

a case-specific approach to data privacy and utility instead of a universal solution regarding synthetic data. For example, the Alan Turing Institute states that universally private and useful synthetic data is infeasible and suggests, instead, tailored assessments for each case (Jordon et al., 2022, p.17).

The British Financial Conduct Authority also emphasizes the importance of evaluating synthetic data, based on its intended application (FCA, 2024). This recommendation extends also to PETs more generally: The European Commission's Joint Research Centre concurs in its 2023 report, noting that no single technique can meet all secure data-sharing requirements (Hurtado Ramírez et al., 2023). In sum, data sharing must be highly tailored and guided by specific criteria for varied scenarios. This suggests that data practitioners potentially need to generate multiple versions of a dataset, depending on the use case. This not only conflicts with the vision of highly versatile data sharing but also does not solve the problem that even seemingly benign data, such as location information from a weather app, can reveal sensitive personal details; these “small privacy risks” can compound and cumulatively still enable invasive inferences (Altman, Cohen and Nissim, 2024).

Secondly, the economic value derived from data sharing hinges on the ability of entities to interconnect diverse data sets. In doing so, new analytical information is formed and can then be converted into knowledge (Kolbe-Guyot and Finger, 2025). The capacity to link, across various data sets, data about a single individual is also crucial for effective data analysis in big-data applications (Soria-Comas and Domingo-Ferrer, 2016, p. 24), as well as in particular sectors such as medical research and generalized health-data sharing (TEHDAS, 2023). Nonetheless, the generation of synthetic data disrupts the direct correspondence between subjects and data measures, thus rendering subsequent linkage with other data sets impossible (Jordon et al., 2022, p.7). This is a significant drawback for data-sharing use cases that are planned for interconnecting diverse datasets via common identifiers. This means that SDG technology is not only unable to eliminate privacy risks, as advertised, but that it also does not yield the anticipated all-purpose utility.

3.3 Market asymmetries in the data supply chain

The recommendations for technical solutions and “silver-bullet” promises in data-sharing also spell difficulties for the burgeoning market for PETs and their intended users, data practitioners. There is an increasing push in the industry to certify certain technologies (such as synthetic data and differential privacy) as providing complete privacy protection, once they are integrated in a system. This idea is attractive to technology providers (i.e., vendors) because it creates a market for their products and to policymakers because they desire straightforward ready-made solutions for everyone to use.

Privacy technologies constitute valuable tools, but viewing them as a universal solution can be overly simplistic and obscure broader systemic responsibilities within the data supply chain. Effective integration of these technologies requires maintaining specific conditions to ensure their effectiveness, a task that demands significant expertise. Data practitioners implementing these technologies, however, often lack this expertise, hence the demand for simple solutions. Additionally, these technologies can affect performance or functionality, which can lead practitioners to make decisions that compromise security, under the mistaken belief that the technology alone will automatically guarantee privacy.²⁶

There is an undesirable market imbalance where technology providers are not accountable for integration mishaps. This places the burden on users who struggle to achieve effective results, without compromising privacy with solutions that are neither straightforward to use nor consistently effective.

Yet, when privacy technologies are certified for privacy protection, any errors are often attributed to those responsible for integrating the technology rather than to the technology itself. Despite the complexity involved in proper implementation (see Stadler, Oprisanu and Troncoso, 2022), technology

²⁶ For example, when configuring differentially private synthetic-data generation algorithms for high privacy, much noise is added, which significantly reduces the data's usefulness. To improve utility, practitioners often use less secure settings. For instance, to reduce noise and make the synthetic data more similar to the original, they extract data ranges from the real data. However, this method introduces real data information into the synthetic data, breaking the rules of differential privacy. As a result, although utility is increased, the synthetic data no longer provides reliable privacy protection for the individuals in the original dataset.

producers, such as those offering synthetic data or differential-privacy tools, frequently make bold claims but assume no liability for any failures. This results in an undesirable market imbalance where technology providers are not held accountable for integration mishaps, thus placing the burden on users who struggle to achieve effective results, without compromising privacy with solutions that are neither straightforward to use nor consistently effective. It is therefore essential to not only put the burden on data practitioners but to establish greater transparency and accountability for the entire data supply chain. If the focus is solely on individual components of the data supply chain, such as privacy-enhancing technologies rather than the entire system, there is a risk of shifting responsibility to tech consumers and allowing tech providers to market their products without guarantees or liabilities.

Moreover, there is currently a lack of a comprehensive data-sharing policy framework that assists practitioners in defining and assessing privacy risks prior to selecting or integrating specific solutions. Key considerations should be firmly rooted in the principles of privacy-by-design. To ensure privacy, careful planning is required throughout the entire system, including the application of privacy technologies. Simply implementing certified privacy tools – without a thorough understanding of their operation and the way they maintain privacy within the overall system – is likely to be insufficient. As previously mentioned, integrating privacy-oriented data-sharing solutions is inherently complex, especially when considering the way the privacy attributes of certain privacy-enhancing technologies can be preserved, given the specific use case and underlying system integration. Yet, despite these complexities, practitioners are often left to navigate these challenges on their own and need greater procedural guidance. Therefore, greater support through the development of the right frameworks is needed.

4. The risks of maintaining a technology-centric approach to data sharing

As outlined above, there are significant gaps between the political vision and objectives for data sharing on the one hand – such as European data spaces, a fair data economy, and enhanced innovation and research – and the technical ability to implement these goals in a privacy-preserving manner on the other. This disconnect is especially pronounced in the context of microdata sharing for general-purpose data uses. Although we agree that PETs must play a crucial role in implementing data-sharing spaces that achieve the desired vision, they cannot fully resolve the inherent tension between broadened data sharing and an increased risk of privacy harms. To meet the demands of data sharing that is both high-utility and privacy-preserving, the use of PETs must be supplemented with enhanced procedural guidance, robust risk-management frameworks, and a paradigm shift in our approach to technology in privacy-sensitive areas.

Continuing to rely solely on technology-centric data-sharing frameworks risks significant consequences, such as information leakage, inaccurate data analysis outcomes, potential future privacy breaches, and increased difficulty for practitioners in assessing privacy risks effectively.

If data-sharing frameworks and architectures continue to be influenced predominantly by current technology-centric policies – those that rely solely on technological solutions for enabling privacy-preserving and high-utility data-sharing scenarios – we will face two significant risks. First, there is an assumption that certain technologies, such as synthetic data, produce privacy-safe data “by design”, i.e., that any dataset produced by these technologies, regardless of implementation details and the specific data use, can be shared with third parties without significant privacy risks. This assumption raises the risk of unintended information leakage. As discussed earlier, data practitioners face considerable implementation challenges and residual privacy risks associated

with SDG, which could lead to disconcerting consequences and privacy harms, depending on the nature and extent of the data involved. Consequently, a one-size-fits-all approach does not suffice. Instead, a case-by-case analysis of the risks is crucial to ensure that the use of PETs indeed provides the desired protections.

Such tailored empirical approaches to data-sharing risk assessment have been advocated by many others. However, their focus has remained on assessing the privacy risks of data sharing and largely overlooked the *utility* aspects of this question (e.g., Altman, Cohen and Nissim, 2024; Jordon et al., 2022). Yet, the data-sharing purpose and its utility requirements determine whether strong privacy guarantees can be provided or if residual risks necessitate procedural controls. For instance, when the objective is to address a specific question or carry out a well-defined analysis, PETs such as federated learning, multi-party computation, or homomorphic encryption, offer robust privacy protections. In these cases, utility is accurately assessed due to well-defined purpose requirements.

Conversely, if the aim is to share data for broad exploratory use, or for big-data applications where purposes can be undefined, procedural controls become essential. This is because extensive utility demands can lead to fundamental information leakage that PETs alone cannot mitigate. Therefore, risk assessments should prioritize the data-sharing purpose, as categorizing use cases by purpose is crucial for determining whether a good trade-off between privacy and utility is feasible.

The data-sharing purpose and its utility requirements determine whether strong privacy guarantees can be provided or whether the residual risks necessitate procedural controls.

As for the second risk, there is the assumption that privacy-preserving data can be analyzed in the same

manner as raw data (original data). This assumption increases the risks of inaccurate conclusions, as the utility for the intended analysis might not be preserved using certain PETs. As previously discussed, SDG does not outperform standard anonymization techniques regarding utility and, in certain data-sharing scenarios, might not be the preferred solution at all. Such assumptions could result in undesirable consequences, particularly in areas such as public policy or consumer analysis. In summary, a reliance on PETs *alone* can enable privacy harms or the harmful inaccuracy of data insights.

Furthermore, as many data practitioners have correctly pointed out, there is a lack of standardized risk-assessment frameworks and regulatory guidelines on privacy risk evaluations for data sharing. Given the complexities involved in balancing appropriate utility and privacy protection – particularly in big-data scenarios where traditional distinctions between PII and non-PII become inadequate – this lack can be especially challenging for small-scale data practitioners.

In contrast, other fields such as medicine, food safety, and environmental protection provide clear guidance and testing procedures to ensure consistent and reliable evaluations. For instance, medical protocols dictate specific standards for patient safety, food safety regulations define acceptable levels of contaminants, and environmental policies set clear limits on pollution levels. This is based on an understanding that the notion of “zero risk” exists only in theory. It thus necessitates careful risk assessments to balance the desirable benefits against the potential downsides.

The same principle should apply to the use of PETs, in particular synthetic data, as suggested, e.g., by the FCA (2023). They correctly argue that “[...] zero risk equates to zero utility”, advocating instead for a transition to risk-based models that outline acceptable risk ranges and provide assessment frameworks that guide data practitioners in adequately estimating privacy risks, such as through individual identification, linkability, inference and motivated intruder threats (FCA, 2023, p.7). However, this is not yet standard in data-sharing, although the first comprehensive guides and tools for conducting, for example, anonymization risk assessments, are available (see Elliot, Mackey and O’Hara, 2020). This increases the likelihood of privacy incidents similar to the re-identification issues seen in the cases of Netflix and AOL.

There is a need for a more comprehensive approach that considers data purpose-driven decision-making that includes risk management frameworks, case-by-case assessments, and clear procedural policies to adjudicate whether privacy or utility should take precedence.

In summary, continuing to rely solely on technology-centric data-sharing frameworks risks significant consequences, such as information leakage, inaccurate data-analysis outcomes, potential privacy breaches, and increased difficulty for practitioners in effectively assessing privacy risks. This underscores the need for a more comprehensive approach that considers purpose-driven decision-making that, to adjudicate whether privacy or utility should take precedence, includes risk management frameworks, case-by-case assessments of privacy-utility trade-offs, and clear procedural policies.

5. Policy recommendations

As the balance between data utility and privacy protection reveals its complexities, a shift from overreliance on technological promises to realistic procedural strategies is essential for creating a more sustainable and trustworthy data-sharing practice. The following recommendations provide steps for moving from a technology-centric paradigm to one that equips policymakers and practitioners to address the privacy-utility trade-off effectively.

→ Recommendation 1

Recognize and accept the inherent trade-off in data sharing

Policymakers and data practitioners should acknowledge the fundamental trade-off between data utility and privacy.

It is essential to acknowledge the fundamental limits of privacy-preserving data sharing. The trade-off between a dataset's utility for statistical analysis and the protection against inference of private information is an intrinsic one, i.e., the trade-off is an inherent and inseparable part of the nature of privacy-preserving data sharing. Any information preserved for analysis can also be misused for harmful inferences about individuals. All data-sharing use cases and any PETs are subject to this *inherent trade-off*.

→ Recommendation 2

Define clearly the purpose of data sharing first

Data-sharing practices should be driven by precise definitions of the data-sharing purpose, before selecting the appropriate PET.

Although purpose limitation is a fundamental principle of data protection, we argue that insufficient emphasis has been placed on precisely *defining* this *purpose*. Current approaches evaluate use cases primarily based on the strengths of the PETs that are employed, i.e., assessing the security and privacy features of the technologies. Whereas, the evaluations should prioritize understanding the purpose, specific considerations, and the objectives behind sharing the data. By prioritizing a clear articulation of the analytical objectives and

detailing the specific outputs, patterns, key independent variables, and methodologies involved, the likelihood of selecting the appropriate PET for the use case is increased.

→ Recommendation 3

Evaluate technical claims critically and realistically

PETs should not be seen as panaceas, as they cannot eliminate entirely the risk of information leakage. Data practitioners should critically evaluate the claims made by technical-solution vendors and avoid accepting blanket promises about privacy-utility balancing technologies.

Data practitioners must critically assess the claims made by vendors of privacy-preserving technologies. Each solution should be evaluated based on two main dimensions: its fit with utility requirements regarding a well-defined purpose and the additional risk it poses compared to fundamental leakage. This requires that every data-sharing use case begins with defining the utility requirements and identifying the baseline fundamental leakage for risk assessments. Then, considerations regarding utility and fundamental leakage should be compared against empirical evidence of each technology's effectiveness in achieving the desired purpose while protecting information. This approach enables a more informed evaluation of technical solutions relative to the established baseline.

→ Recommendation 4

Develop standards to evaluate the inherent trade-off

Instead of over-relying on future technological solutions, policymakers should prioritize the development of guidance in the form of standards that determine the prioritization of privacy or utility.

To facilitate better decision-making for data practitioners, the development of procedural standards and policies to evaluate trade-offs between utility preservation and privacy risks is further recommended. These standards and policies should clarify relevant legal and regulatory aspects, guide the decision-making process through purpose definitions and risk assessment, and help to determine the best use of privacy-enhancing technologies by providing guidance on evaluation methodologies. Current efforts for developing privacy-preserving standards, such as those by the IEEE dedicated standards committee, are important but focus primarily on technology management; alone, they do not adequately assist data owners in navigating the utility-privacy trade-offs. Future policies need to go beyond technical and management standards to also include procedural standards to *help evaluate trade-offs inherent to any data-sharing use case*.

→ Recommendation 5

Shift to purpose-led, risk-based approach to data sharing

Policymakers should support the establishment of a purpose-led, risk-based approach to data-sharing that evaluates and manages inherent risks, based on the specific objectives of data use.

When data is used for specific queries, privacy-enhancing technologies can effectively ensure privacy due to clearly defined requirements. Whereas, for data meant to be broadly applicable and for big-data scenarios, where purposes can be undefined, data sharing cannot exist without risks. For those cases, as advocated by others,²⁷ this paper recommends shifting to a risk-based approach to data sharing; an approach that assesses and manages the inherent risks associated with data-sharing scenarios and technologies. We also recommend that a risk-based approach to data-sharing should be led by the purpose of the data-sharing use case. The

purpose of data sharing and its utility requirements dictate whether PETs can ensure strong privacy guarantees or if residual risk remains and procedural controls are needed. Therefore, to better manage the privacy-utility trade-off, future policies should prioritize defining the data-sharing purpose.

→ Recommendation 6

Encourage tailored approaches

Practitioners should be encouraged to evaluate privacy technologies in the context of specific data use cases by considering their purpose and utility requirements.

As advocated by others²⁸, we recommend that, rather than a one-size-fits-all solution, data-sharing considerations should involve tailored approaches and case-specific assessments to select the appropriate PET. We add to this that, instead of focusing on guarding against every hypothetical or theoretical risk of identifiability, the focus should be on assessing the likelihood of individual identification, given the needed utility requirements. For this, data practitioners should group potential use-cases by the types of utility and features they require and evaluate a PET's performance, based on the data-sharing use case's intended application.

→ Recommendation 7

Enhance transparency and accountability in the data-sharing supply chain

Policymakers should enhance transparency and accountability in the data-sharing supply chain by requiring vendors to disclose technology blind spots and risk strategies, thus encouraging the sharing of privacy responsibilities between vendors and data owners while building trust.

Currently, the onus of ensuring the right privacy-protections is placed on data owners. To foster a robust and secure data-sharing ecosystem, it is important, however, to encourage transparency and improve accountability throughout the data supply chain. To this end, vendors should be required to provide greater transparency regarding the blind spots of the technologies and their strategies for managing them. This should include acknowledging inherent risks regarding

²⁷ See e.g. FCA (2024), Elliot, Mackey and O'Hara (2020), Citron and Solove (2001).

²⁸ See e.g. Altman, Cohen and Nissim (2024) and Jordon et al. (2022).

both private information leakage, as well as a loss in data utility. Demanding improved transparency (visible to data practitioners) can ensure that the technologies used are not only effective but also align with ethical and legal standards.

→ Recommendation 8

Support ongoing dialogue and research

To ensure that policies are informed by the latest empirical research and technological developments, policymakers and data practitioners should remain engaged in continuous dialogue with researchers about the evolving capabilities and limitations of PETs.

By fostering regular interactions among researchers, technology developers, and regulatory bodies, stakeholders can collaboratively address emerging challenges and insights. Being informed enables policymakers to craft adaptive policies that reflect new developments, and practitioners can implement solutions that meet both legal and ethical standards. This approach helps align political objectives with technical capabilities, thus promoting responsible and safe data-sharing practices.

6. Conclusion

In this paper, we have highlighted the problematic reliance on PETs for data sharing as “silver bullet” solutions for data sharing in privacy-sensitive contexts. Through a detailed analysis, including the rise of SDG as a case study, we have argued that technical solutions are important but alone cannot resolve the inherent trade-off between preserving utility and guaranteeing privacy. This paper stresses, instead, the necessity for realistic policy approaches that recognize these limitations. This is especially relevant in the context of EU ambitions for establishing sectoral data spaces and for facilitating widespread (micro)data sharing. This paper calls for an informed and tailored approach to data-sharing policy among policymakers and data practitioners. It urges a shift from a technology-centric approach to one that openly recognizes fundamental trade-offs and that prioritizes thorough evaluations centered around the intended purpose of the data sharing, as well as transparency across the data supply chain.

In doing so, this paper seeks to offer clearer guidance for data practitioners in evaluating the purposes, needs, and risks associated with data sharing. It is not an endorsement for increased intervention through committee-based management approaches, which could lead to excessive bureaucracy and hinder effective data sharing. Instead, we advocate for a balanced approach – one that equips data practitioners with straightforward guidelines and adaptable policy frameworks. This approach should emphasize a deep understanding of specific data-sharing objectives, careful risk assessment, and the implementation of suitable safeguards, all while avoiding unnecessary bureaucratic layers.

Likewise, this paper maintains that data-sharing PETs play a crucial role in implementing privacy-preserving data sharing. Their adequacy depends on the specific goals of the data use, however. PETs like federated learning, multi-party computation, or homomorphic encryption offer strong privacy protections when the intended data use is to answer well-defined questions or perform specific analyses. In these cases, technological solutions might successfully minimize privacy risks while providing the desired data utility

because intended purpose and utility needs are clearly defined. However, when data is to be shared for broad exploratory purposes, such as in the case of data spaces, the fundamental trade-off between preserving information for valid statistical analysis and preventing harmful inferences implies that PETs alone might not simultaneously achieve high utility and high privacy. In such cases, procedural controls are needed to adequately manage this trade-off.

All in all, this paper underscores the importance of recognizing the limitations of data-sharing technologies from a policy standpoint. As policy circles and the EU Commission explore the potential development of large-scale data sharing with privacy-preserving methods, there is an urgent need for a realistic and technologically grounded assessment of the data-sharing technologies proposed to achieve this vision. The paper calls for a more critical evaluation of these technologies, emphasizing the need for greater scientific and technical input to guide informed and effective policy decisions.

References

- Ada Lovelace Institute (2022). *Rethinking data and rebalancing digital power*. London: Ada Lovelace Institute. Available at adalovelaceinstitute.org/report/rethinking-data/ [accessed: 16.05.2025].
- Altman, M., Cohen, A. and Nissim, K. (2024). "ACM tech brief: data privacy protection." *ACM Technology Policy Council*, 11(Spring), pp. 1–4.
- Annamalai, M. S., Gadotti, A. and Rocher, L. (2024). "A linear reconstruction approach for attribute inference attacks against synthetic data." In *Proceedings of the 33rd USENIX Security Symposium*, pp. 2351–2368
- Article 29 Data Protection Working Party (2014). *Opinion 05/2014 on anonymisation techniques*. 0829/14/EN WP216. Available at ec.europa.eu/justice/article-29/documentation/opinion-recommendation/files/2014/wp216_en.pdf [accessed: 26.11.2024].
- Bellovin, S.M., Dutta, P.K. and Reiter, N. (2019). "Privacy and synthetic datasets." *Stanford Technology Law Review*, 22(1), pp. 1–52.
- Citron, D.K. and Solove, D.J. (2021). "Privacy harms." *Boston University Law Review*, 102(3), pp. 793–863.
- Cramer, R., Damgard, I.B. and Nielsen, J.B. (2015). "Introduction." In *Secure multiparty computation and secret sharing*. Cambridge: Cambridge University Press, pp. 3–13.
- De Montjoye, Y.-A., Hidalgo, C.A., Verleysen, M. and Blondel, V.D. (2013). "Unique in the crowd: the privacy bounds of human mobility." *Scientific Reports*, 3(1376), pp. 1–5.
- Dinur, N. and Nissim, K., 2003. "Revealing information while preserving privacy." In *Proceedings of the 22nd ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*, pp. 202–210. Available at dl.acm.org/doi/10.1145/773153.773173 [accessed: 26.11.2024].
- Drechsler, J. (2011). *Synthetic datasets for statistical disclosure control: theory and implementation*. New York: Springer.
- Dwork, C. and Roth, A. (2014). "The algorithmic foundations of differential privacy." *Foundations and Trends in Theoretical Computer Science*, 9(3–4), pp. 211–407.
- Dwork, C., Naor, M., Reingold, O., Rothblum, G.N. and Vadhan, S. (2009). "On the complexity of differentially private data release: efficient algorithms and hardness results." In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing*, pp. 381–390.
- Elliot, M., Mackey, E. and O'Hara, K. (2020). *The anonymisation decision-making framework: European practitioners guide*. 2nd ed. Manchester: UK Anonymisation Network UKAN. Available at ukanon.net/framework/ [accessed: 01.05.2025].
- European Commission (2020). *A European strategy for data*. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, COM(2020) 66 final, 19 February. Brussels: European Commission. Available at eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52020DC0066 [accessed: 26.11.2024].
- European Commission (2022). *Commission staff working document on common European data spaces*, SWD(2022) 45 final. Brussels: European Commission. Available at ec.europa.eu/newsroom/dae/redirection/document/83562 [accessed: 27.11.2024].
- European Commission (2023). *Commission implementing Regulation (EU) 2023/138 of 21 December 2022 laying down a list of specific high-value datasets and the arrangements for their publication and re-use*. Brussels: Official Journal of the European Union, L 19, pp. 43–75. Available at eur-lex.europa.eu/eli/reg/impl/2023/138/oj [accessed: 26.11.2024].
- European Union (2022a). *Regulation (EU) 2022/868 of the European Parliament and of the Council of 30 May 2022 on European data governance and amending Regulation (EU) 2018/1724 (Data Governance Act)*. Brussels: Official Journal of the European Union, L 152, pp. 1–44. Available at eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:3A32022R0868 [accessed: 26.11.2024].
- European Union (2022b). *Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on contestable and fair markets in the digital sector and amending Directives (EU) 2019/1937 and (EU) 2020/1828 (Digital Markets Act)*. Brussels: Official Journal of the European Union, L 265, pp. 1–66. Available at data.europa.eu/eli/reg/2022/1925/oj [accessed: 26.11.2024].
- European Union (2023). *Regulation (EU) 2023/2854 of the European Parliament and of the Council of 13 December 2023 on harmonized rules on fair access to and use of data and amending Regulation (EU) 2017/2394 and Directive (EU) 2020/1828 (Data Act)*. Brussels: Official Journal of the European Union, OJ L, pp. 1–71. Available at eur-lex.europa.eu/eli/reg/2023/2854/oj/eng [accessed: 26.11.2024].
- European Union (2025). *European approach to artificial intelligence*. Available at digital-strategy.ec.europa.eu/en/policies/european-approach-artificial-intelligence [accessed: 25.11.2024].
- Farrall, F., Mittal, N., Narra, C., Tello, J. and Dow, E. (2022). *Data-sharing made easy: powerful data-sharing and privacy-preserving technologies usher in a new era of data monetization*. Deloitte Insights. Available at www2.deloitte.com/us/en/insights/focus/tech-trends/2022/data-sharing-technologies.html [accessed: 26.11.2024].
- Farrell, E., Minghini, M., Kotsev, A., Soler-Garrido, J., Tapsall, B., Micheli, M., Posada, M., Signorelli, S., Tartaro, A., Bernal, J., Vespe, M., Di Leo, M., Carballa-Smichowski, B., Smith, R., Schade, S., Pogorzelska K., Gabrielli, L. and de Marchi, D. (2023). *European data spaces: scientific insights into data sharing and utilisation at scale*. Luxembourg: Publications Office of the European Union.
- FCA (2023). *Exploring synthetic data validation – privacy, utility and fidelity*. London: UK Financial Conduct Authority. Available at www.fca.org.uk/publications/research-articles/exploring-synthetic-data-validation-privacy-utility-fidelity [accessed: 27.11.2024].
- FCA (2024). *Using synthetic data in financial services: the synthetic data expert group*. London: UK Financial Conduct Authority. Available at www.fca.org.uk/publications/corporate-documents/report-using-synthetic-data-financial-services [accessed: 26.11.2024].
- Gadotti, A., Rocher, L., Houssiau, F., Cretu, A.-M. and de Montjoye, Y.-A. (2024). "Anonymization: the imperfect science of using data while preserving privacy." *Science Advances*, 10(29), pp. 1–22.

Ganev, G., Oprisanu, B. and de Cristofaro, E. (2022). "Robin Hood and Matthew effects: differential privacy has disparate impact on synthetic data." In *Proceedings of the 39th International Conference on Machine Learning (ICML 2022)*, PMLR 162, pp. 6944–6959.

Granel, C., Mooney, P., Jirka, S., Rieke, M., Ostermann, F., Van Den Broecke, J., Sarretta, A., Verhulst, S., Dencik, L., Oost, H., Micheli, M., Minghini, M., Kotsev, A. and Schade, S. (2022). *Emerging approaches for data-driven innovation in Europe: sandbox experiments on the governance of data and technology*. Luxembourg: Publications Office of the European Union.

Hradec, J., Craglia, M., Di Leo, M., De Nigris, S., Ostlaender, N. and Nicholson, N. (2022). *Multipurpose synthetic population for policy applications*, EUR 31116 EN. Luxembourg: Publications Office of the European Union.

IAPP (2010). *Synthetic data offers advanced privacy for the census bureau, business*. Available at iapp.org/news/a/synthetic-data-offers-advanced-privacy-for-the-census-bureau-business/ [accessed: 01.05.2025].

Hurtado Ramírez, D., Porras Díaz, L., Rahimian, S., Auñón García, J.M., Irigoyen Peña, B., Al-Khazraji, Y., Gavín Alarcón, Á., González Fuente, P., Soler Garrido, J. and Kotsev, A. (2023). *Technological enablers for privacy-preserving data sharing and analysis*. Luxembourg: Publications Office of the European Union.

IEEE (2023). *Standards association industry connections activity "synthetic data" (IC21-013)*. Available at standards.ieee.org/industry-connections/synthetic-data/ [accessed: 01.05.2025].

IBM (2020). *Smart manufacturing: AI technologies, intelligent insights*. Expert Insights. Armonk, NY: IBM. Available at www.ibm.com/thought-leadership/institute-business-value/en-us/report/smart-manufacturing [accessed: 16.05.2025].

Frost and Sullivan (2019). *Industry 4.0, the fourth revolution: challenges, benefits, adoption and how to begin*. Available at www.ibm.com/topics/industry-4-0 [accessed: 16.05.2025].

Jordon, J., Houssiau, F., Cherubin, G., Cohen, S., Szpruch, L., Bottarelli, M., Maple, C. and Weller, A. (2022). "Synthetic data: what, why and how?" Commissioned by the Royal Society. London: The Alan Turing Institute. Available at arxiv.org/abs/2205.03257 [accessed: 26.11.2024].

Joint Research Center, 2024. *Digital financial services: synthetic*

data ensures compliance with confidentiality requirements. Available at joint-research-centre.ec.europa.eu/jrc-news-and-updates/digital-financial-services-synthetic-data-ensures-compliance-confidentiality-requirements-2024-03-20_en [accessed: 01.05.2025].

Kolbe-Guyot, M. and Finger, M. (2025). *Datenstrategie, -Policy und -Regulierung für die Schweiz*. C4DT Insight #2. Lausanne: Center for Digital Trust (C4DT), EPFL. Available at c4dt.epfl.ch/article/?cat=61 [accessed: 01.05.2025].

Lu, P.-H., Wang, P.-C. and Yu, C.-M. (2019). "Empirical evaluation on synthetic data generation with generative adversarial network." In *Proceedings of the 9th International Conference on Web Intelligence, Mining and Semantics*. Association for Computing Machinery, pp. 1–6.

Meeus, M., Guépin, F., Cretu, A.-M. and de Montjoye, Y.-A. (2023). "Achilles' heels: vulnerable record identification in synthetic data publishing." In: Tsudik, G., Conti, M., Liang, K., and Smaragdakis, G., eds., *Computer Security – ESORICS 2023. Lecture Notes in Computer Science*, 14345. Cham: Springer Nature Switzerland, pp. 380–399.

MostlyAI (2025). *Data sharing: fix broken data access with synthetic data*. Available at mostly.ai/use-case/data-democratization [accessed: 01.05.2025].

Narayanan, A. and Shmatikov, V. (2008). "Robust de-anonymization of large sparse datasets." In *Proceedings of the 2008 IEEE Symposium on Security and Privacy*, pp. 111–125.

Narayanan, A. and Shmatikov, V. (2010). "Privacy and security: myths and fallacies of 'personally identifiable information.'" *Communications of the ACM*, 53(6), pp. 24–26.

Ohm, P. (2010). "Broken promises of privacy: Responding to the surprising failure of anonymization." *UCLA Law Review*, 57, pp. 1701 – 1777.

Schneider, M. (2021). *Census bureau's use of 'synthetic data' worries researchers*. Associated Press, 27 May. Available at apnews.com/article/census-2020-technology-data-privacy-business-be938fa5db887a0ae6858dff0be217ef [accessed: 01.05.2025].

Solove, D. J. (2024). "Data is what data does: regulating based on harm and risk instead of sensitive data." *Northwestern University Law Review*, 118 (4), pp. 1081–1138.

Soria-Comas, J. and Domingo-Ferrer, J. (2016). "Big data privacy: challenges to privacy principles and models." *Data Science and Engineering*, 1(1), pp. 21–28.

Stadler, T., Oprisanu, B. and Troncoso, C. (2022). "Synthetic data: anonymisation groundhog day." In *Proceedings of the 31st USENIX Security Symposium*, pp. 1451–1468.

Sweeney, L. (2000). "Simple demographics often identify people uniquely." *Health*, 671, pp. 1–34.

Sweeney, L., Loewenfeldt, M. and Perry, M. (2018). "Saying it's anonymous doesn't make it so: re-identifications of 'anonymized' law school data." *Technology Science* (12 November). Available at techscience.org/a/2018111301/ [accessed: 01.05.2025].

Swiss Federal Assembly (2020). *Federal Act on Data Protection* (Data Protection Act, FADP). Available at www.fedlex.admin.ch/eli/cc/2022/491/en [accessed: 01.05.2025].

Swiss Federal Council (2022). *Promotion of trustworthy data spaces and digital self-determination*. Press release, 30 March. Available at www.news.admin.ch/en/nsb?id=87780 [accessed: 21.04.2025].

SynthIA, 2025. *Leverage synthetic data to propel personalized medicine to new heights*. Available at www.ih-synthia.eu [accessed: 01.05.2025].

TEHDAS Joint Action (2023). *Recommendations on a data quality framework for the European health data space for secondary use*. Helsinki: Finnish Innovation Fund (SITRA). Available at tehdas.eu/app/uploads/2023/09/tehdas-recommendations-on-a-data-quality-framework.pdf [accessed: 26.11.2024].

VentureRadar (2022). *Top privacy preserving analytics companies*. Available at www.ventureradar.com/keyword/Privacy%20Preserving%20Analytics [accessed: 26.11.2024].

Ullman, J. and Vadhan, S. (2011). "PCPs and the hardness of generating private synthetic data." In *Theory of Cryptography Conference*. Berlin, Heidelberg: Springer, pp. 400–416.

Wood, S., Wilkinson, L., Kaur, A., Schoentgen, A. and Lavender, T. (2022). *Review of literature relevant to data protection harms*. London: UK Information Commissioner's Office. Available at ico.org.uk/media/2/about-the-ico/documents/4020142/plum-review-of-literature-relevant-to-data-protection-harms-v1-202203.pdf [accessed: 16.05.2024].

Wood, A., Altman, M., Bembenek, A., Bun, M., Gaboardi, M., Honaker, J., Nissim, K., O'Brien, D. R., Steinke, T. and Vadhan, S. (2020). "Differential privacy: a primer for a non-technical audience." *Vanderbilt Journal of Entertainment and Technology Law*, 21(209), pp.209–276.

Yale, A., S. Dash, R. Dutta, I. Guyon, A. Pavao and Bennett, K. P. (2019). "Assessing privacy and quality of synthetic health data." In *Proceedings of the Conference on Artificial Intelligence for Data Discovery and Reuse*, pp.1–4.

Yao, Z., Krčo, N., Ganev, G. and de Montjoye, Y.-A. (2025). "The DCR delusion: measuring the privacy risk of synthetic data." arXiv preprint, arXiv:2505.01524. Available at arxiv.org/abs/2505.01524 [accessed: 16.05.2024].

Yin, X., Zhu, Y. and Hu, J. (2021). "A comprehensive survey of privacy-preserving federated learning: a taxonomy, review, and future directions." *ACM Computing Surveys*, 54(6), pp. 4–6.

About

C4DT Insight

C4DT Insight publications are extensive expert reports and studies that scientifically explore issues at the intersection between technology and policy; they contain substantiated political recommendations for action, and contribute to evidence-based policy making.

Center for Digital Trust

Housed at the Swiss Federal Institute of Technology Lausanne (EPFL, epfl.ch), the Center for Digital Trust (C4DT, c4dt.epfl.ch) brings together academy, industry, not-for profit organizations, civil society, and policy actors to collaborate, share insight, and to gain early access to trust-building technologies by relying on state-of-the-art research at EPFL. C4DT supports the public sector by acting as an expert and by facilitating technology transfer in domains such as privacy protection and security, democracy and humanitarian assistance, and critical infrastructures.

EPFL C4DT
Building BC
Station 14
1015 Lausanne
Switzerland

About the authors

Dr Theresa Stadler is a postdoctoral researcher at the Security and Privacy Engineering Lab at EPFL.

Prof. Dr. Carmela Troncoso is a scientific director at the Max Planck Institute for Security & Privacy, where she heads the Security and Privacy Engineering Lab.

Dr Melanie Kolbe-Guyot is head of policy at the Center for Digital Trust (C4DT) at EPFL and tech policy fellow at the University of California, Berkeley.

The views expressed in this publication are not necessarily those of the C4DT. Any commercial use of the media published by C4DT without the written consent of C4DT is not permitted.